Low-Power Meteor Detection System

Community Partner: Rose City Astronomers (RCA)
Contact Name: Mark Martin
Contact email: marmart2@pdx.edu

IP and Compliance Requirements

- Students must sign an NDA: No
- Intellectual Property will be owned by: Negotiable, Intending to Open-Source
- Students must be US persons: No
- Other legal requirements (background checks, drug testing, etc): None

Background and Motivation

The **Rose City Astronomers (RCA)** is a nearly 1000-family (roughly 2000-person) all-volunteer, non-profit astronomy club with close ties to the **Oregon Museum of Science and Industry (OMSI)**. One way RCA supports OMSI is that we provide the telescopes and volunteers to staff the OMSI public star parties. A star party is a gathering of people at night with telescopes and public star parties are events where people share views through their telescopes with the public.

Theory:

At several public star parties, members of our Radio Astronomy Special Interest Group have set up forward-scatter meteor detection systems. This type of system consists of one or more radio receivers tuned to the frequency of a strong transmitter located over the horizon from the receivers. Since the transmitter is hidden from the receivers by the curvature of the Earth, a receiver only receives signals from the transmitter that have been reflected off objects that are in the line of sight of both the transmitter and the receiver. (See Figure 1.) Most often, the received signals are reflected off the trails of meteors. These trails are created when a meteor enters the atmosphere and its outer layers burn up and become ionized. (Most meteors are very small and the entire meteor burns up.) The frequencies observed at the star parties are associated with two television transmitters – one in Los Angeles and another in Las Vegas.

Meteor detectors have been very popular at star parties. People enjoy seeing meteors that systems detect and also enjoy expressing skepticism about detectors when a signal is detected but a meteor isn't visible. This has generated additional interest in meteor detection systems within the RCA Radio Astronomy SIG. As it turns out, radio-based meteor detectors are popular in Japan, where people have been using multiple receivers to locate meteors and sometimes use the tracking data to search for the meteoroids from larger meteors that didn't entirely burn

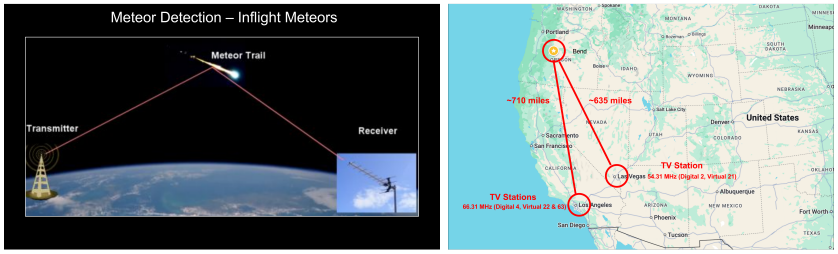


Figure 1: Forward-Scatter Meteor Detection System Used at the OMSI Perseid Meteor Shower Public Star Party

up in the atmosphere. (For example, see Exp Astron (2016) 41:243–257, DOI 10.1007/s10686-015-9488-4.) These detectors can also be operated during the day, making them easier on people's circadian rhythms than most astronomical activities.

Research Contribution:

Aside from the "Wow!" factor, the fun of debating the validity of a detection, and the possibility of tracking meteors and hunting meteoroids, this type of system can be used to accomplish some scientific goals. The strength of a reflection can be used to estimate the size of a meteor and a set of measurements can be used to characterize the population of meteors entering the atmosphere within range of the detector. How the signal changes when it is reflected off of a meteor's trail can also carry clues regarding the chemical composition of the meteor. For some of these purposes, it is necessary to control the transmitter as well as the receivers. If a receiver is in the line of sight of the transmitter, the signals received might be back-scattered from the targets, as in a radar system, rather than forward-scattered.

Transmitter <50W, Technician license

A transmitter adequate for a meteor detection system can operate at surprisingly low powers, reducing FCC licensing requirements. For example, the transmitter described in the above paper operates at 50 W. Such a transmitter paired with multiple receivers that are sufficiently far apart and one or more all-sky cameras can achieve all of the above objectives and provide additional optical information and confirmation for bright meteors.

UniCLOGS status TBD

The Portland State Aerospace Society (PSAS) has very graciously offered to allow us to install our equipment at their soon to be upgraded University Class Open Source Ground Station (UniClOGS), which will be mounted on top of the Vernier Science Center. We hope to connect the system to the university network so it can be remotely operated on campus.

The **Society of Amateur Radio Astronomers (SARA)** is working toward establishing a meteor detection network that covers all of North America. This project might be able to collaborate with this effort or establish complementary resources that can play a role in the SARA network. Along with experts and support from RCA, SARA is also likely to be an excellent resource for the

Industry/Academic resources

current project and has pioneered equipment that is directly relevant. The project will be able to leverage equipment from the Radio and Radio Astronomy at Portland State ([RA]^{2}PS) club. In addition, there are professors in the Physics Department at Portland State who specialize in meteors and I'm hoping they will be willing to consult for the project.

Objective

The goals of this project are to

1. purchase the components of a low-power meteor detection system and set up the system on top of the Vernier Science Center, as part of UniClOGS,
2. connect the system to the university network so that it is accessible and operable from the [RA]^{2}PS lab in the FAB,
3. learn how to process the data from multiple receivers to track and possibly analyze meteors,
4. work with RCA members who have their own equipment to include their receivers in the detection network,
5. work with members of RCA to make pilot measurements and apply techniques for characterizing meteors.

[RA]^{2}PS ("Radio and Radio Astronomy at Portland State") is a student club that was formed in the ECE department last year with faculty advisors me (marmart2@pdx.edu) and Josh Mendez (joshmen@pdx.edu). This project is intended to contribute to the activities of [RA]^{2}PS and the RCA Radio Astronomy SIG. It will create a prototype that could be used in additional locations across the region and/or be connected to the SARA network.

The complete system will consist of

- a low-power transmitter, <50W
- one or more receiving antennas,
- one or more software defined radios (SDRs) that receive signals from the antennas and convert them into digital signals,
- an all-sky camera to create optical images of any detections,
- a laptop or single-board computers (SBCs) running open-source software that processes signals from the SDRs.

It would be best to build an initial system consisting of

- the transmitter,
- 1 receiving antenna,
- 1 SDR,

- the all-sky camera,
- and a laptop or SBC.

Initial System Goal

It would be best to connect the initial system to the university network and learn how to make detections with it before adding additional antennas, SDRs, and computers. It will probably be necessary to determine appropriate custom settings for calibrating and operating the SDR and all-sky camera. Also, ways to power the system and the overall budget are yet to be determined.

Budget???

Requirements

The team

Must

- Purchase and build the simplest initial system, as described above. This will require
  - choosing components to match the objectives and fit a budget that is currently TBD,
  - determining how to power the system.
- Connect the simplest system to the university network so it is accessible from the RAPS lab. This means it will be possible to
  - interactively view detections from the lab,
  - control the SDR from the lab,
  - control the all-sky camera from the lab and collect resulting images.
- Add (at least) two additional receiving antennas to the detection network.
- Create software for using data from 3 or more receivers to track meteors.

Should

- Attempt to connect at least one remote receiver that an RCA member owns to the network or at least determine how to incorporate data collected by a remote receiver in the analysis of meteors.

May

- Explore other techniques for characterizing meteors. These might involve creating custom signals to transmit.

Suggested Skills

Important skills for the team to have coming into the project:

- Knowledge of signal processing,
- Experience connecting devices via the Internet,
- Intermediate software development skills.

Skills that would be helpful for the team but can be acquired during the project:

- Experience with applications of radio, antennas, and SDRs,
- Understanding digital cameras and their settings,

- Familiarity with Linux and open-source software.

Deliverables

Beyond the Project Proposal and a Final Report, the team must provide

- A demonstration showing operation of the detection network.
- A short presentation (roughly 25 minutes) to the May 17th, 2027 meeting of RCA describing the work completed. (This will essentially be a celebration of the team's accomplishments.)
- A user manual describing how to operate the detector.
- A developer's manual describing details of the implementation, internal limitations, and future tasks.

There may be tasks remaining at the end of the project. These could provide opportunities for additional work past the end of the project. There might also be additional opportunities for presenting the work of the team in various forums.